# Self-Healing Routing on a Real Road Network: Ant Colony Optimization and Physarum versus Classical Shortest-Path Algorithms

K Karthik Pai
*Dept. of Computer Science and Engineering*
*RV College of Engineering*
Bengaluru, India
1RV25CS076

Karan Koder
*Dept. of Computer Science and Engineering*
*RV College of Engineering*
Bengaluru, India
1RV25CS080

Kushal Gowda S
*Dept. of Computer Science and Engineering*
*RV College of Engineering*
Bengaluru, India
1RV25CS090

N Litheesh
*Dept. of Computer Science and Engineering*
*RV College of Engineering*
Bengaluru, India
1RV25CS107

Dr. Lingayya Hiremath
*Dept. of Biotechnology*
*RV College of Engineering*
Bengaluru, India
Faculty Mentor

***Abstract*—Bio-inspired routing methods are claimed to *self-heal*: to adapt from retained internal state after a disruption rather than recompute. That claim is usually evaluated on synthetic undirected graphs without congestion, and rarely by measuring recovery. We benchmark Dijkstra/A*, a MAX–MIN Ant System, and a Physarum conductivity model on the real road network of central Bengaluru (2,856 nodes, 7,015 directed edges, mean out-degree 2.46) under SUMO-simulated congestion, across seven closure scenarios and warm- versus cold-start recovery: 5,520 runs, fully paired. We report two results. First, *path extraction decides what a Physarum number means*: pooled, the model matches the optimum on 90% of undisrupted pairs, but using only its own maximum-flux walk the figure is 23.3%, and on warm recoveries it returns the Dijkstra fallback's path on 66.8% of cells; the median stretch is exactly 1.000 inside every provenance class, so a pooled figure cannot be falsified. Second, *warm start is a trade*: it cuts recovery 2.97× in iterations and 6.87× in wall-clock ($n = 358$, $p < 10^{-51}$), with 48.3% of warm recoveries converging in one iteration where cold start never does, yet it is significantly worse on the reported path ($p = 4.3 \times 10^{-7}$). We argue the sparse directed structure of a real road network is what exposes these effects, and that they are structurally unreachable on the complete and near-complete undirected graphs used in prior evaluations.**

***Index Terms*—Bio-inspired routing, ant colony optimization, Physarum polycephalum, network resilience, traffic simulation, benchmarking.**

## I. Introduction

Ant colony optimization (ACO) [1], [2] and Physarum polycephalum conductivity models [3], [4] maintain internal state (a pheromone field, a field of tube conductivities) that is claimed to let them adapt to a changed network rather than recompute from scratch. On a road network, where closures and congestion change edge costs continuously, that is an attractive property.

It is also rarely measured. Evaluations of these methods overwhelmingly use synthetic graphs that are undirected and comparatively dense, report accuracy on an undisrupted shortest-path problem, and treat the adaptive machinery as self-evidently useful. Three gaps follow: the test graphs lack the structure that makes real road routing hard (one-way streets, low out-degree, cul-de-sacs); congestion is usually absent, so the weights the adaptation tracks never move; and recovery after disruption, the behaviour the retained state exists to provide, is seldom the measured quantity.

We close those gaps for one city. We benchmark Dijkstra and A* [5], [6], a MAX–MIN Ant System, and a Tero-style Physarum model on the real road network of central Bengaluru under congestion from the SUMO microsimulator [7], measuring recovery from seven classes of closure under warm start (retained state) and cold start (state reset). Every algorithm sees identical OD pairs, closures and traffic, and all quality figures use a Dijkstra optimum computed on the same weighted graph.

**Contribution 1: extraction provenance, not the model, decides what a Physarum result means.** The model emits a field, not a route; a separate extraction step produces the path. Instrumenting it shows Physarum's apparent competence is largely its fallback's: pooled, it matches the optimum on 90.0%

of undisrupted pairs, reproducing the literature; restricted to its own maximum-flux walk, 23.3%. The median stretch is exactly 1.000 in *every* provenance class, so a pooled number cannot distinguish the model from its Dijkstra fallback, which supplies 66.8% of reported paths on warm recoveries.

**Contribution 2: warm start buys speed and charges quality.** It cuts recovery 2.97× in iterations and 6.87× in wall-clock, paired ($n = 358$), and 48.3% of warm recoveries converge in exactly one iteration, which cold start never does, at unchanged per-iteration cost: the saving is adaptation, not cheaper arithmetic. Against this, warm start is significantly worse on the reported post-disruption path ($p = 4.3 \times 10^{-7}$) and completes its own extraction far less often.

We report what the methods lose as well as gain: Dijkstra recomputes an exact route in one iteration and about 31 ms, which no method here approaches.

## II. Related Work

**Physarum shortest path.** The slime mould resolves a maze into the shortest connecting path [3]; Tero et al. formalised this as adapting conductivities driven by a Poiseuille-flow pressure solve [4], later used to reproduce Tokyo rail-like networks [8]. Bonifaci et al. proved convergence to the shortest path for *undirected* graphs [9]. The directedness restriction is explicit in the literature: Zhang et al. note the model is designed only for undirected networks and propose a directed formulation [10], which we do not implement. Our results therefore characterise the canonical model on a directed network, an acknowledged limitation, rather than reporting a defect.

**The prior three-way comparison.** Combining Physarum, ACO and Dijkstra is not new. Zhang et al. benchmark exactly these three [11], reporting that their Physarum variant and Dijkstra both attain optimal paths "with one hundred percent" while ACO degrades with network size, on Erdős–Rényi graphs of 15–2,000 nodes with static weights, no congestion and no disruption. Two points matter. Their method needs *no* extraction machinery, since the path is read straight off the flux field, which separates cleanly on such graphs, so we cannot claim that Physarum evaluations generally conflate provenance; the honest contrast is that extraction is a formality there and decisive here. And their ACO reaches zero optimal-path accuracy beyond 1,000 nodes, whereas ours attains 43–44% exact match on a larger and harder network, which establishes our ACO as a strong baseline by the standard of the prior art.

**Extraction and termination.** Wang et al. define a dominant-path procedure procedurally identical to our maximum-flux walk [12], and state that at convergence it yields exactly the shortest path. On our network it yields a directed-legal path on 43.0% of solves and the optimum on 23.3%. The results are compatible: their examples are complete graphs and small-worlds of mean degree 6–600, assumed undirected, where the walk *cannot* dead-end. Our network has mean out-degree 2.46 with pervasive one-ways.

**ACO on traffic networks.** Zhang et al. propose orientation guidance weighting edges by angular alignment with the destination [13], evaluated on one 528-node network with static travel times against conventional ACO only; our goal-directed ablation is related but distinct and we extend rather than contradict it. Tuan et al. report an improved ACO beating conventional ACO, Dijkstra and a genetic algorithm [14], but evaluate on METR-LA under fluctuating traffic [15] without measuring recovery after a closure.

**Position.** A systematic pass over the open citation databases, including complete forward-citation sweeps of both [11] and [16], returns no study combining the three families on a real road network under simulator-generated congestion with closure recovery as a measured outcome; no work pairing Physarum with SUMO was found at all. The contribution claimed here is the substrate and the measurement, not the three-algorithm combination, which [11] already reported.

## III. Method

### A. Network and congestion

We study a ∼4×4 km area of central Bengaluru (MG Road, Cubbon Park, Shivajinagar, Richmond Town), chosen for its irregular organically grown topology. The network comes from OpenStreetMap via OSMnx [17], reduced to its largest strongly connected component: **2,856 nodes, 7,015 directed edges, mean out-degree 2.46**. That last figure is the decisive structural property: a node with out-degree two offers a walk-based extraction almost no choice, and one-way streets mean a direction favoured by an undirected flow computation may be illegal.

Algorithms never see the congestion source directly: a provider returns a snapshot mapping each edge to a congested travel time. The primary source is SUMO [7] at light and heavy synthetic demand, joined to graph edges through OSM way IDs at **97.53% coverage**. An analytic BPR volume-delay function [18] is retained as an independent secondary source.

Two details affect correctness. Because `netconvert` splits ways that OSMnx merges, per-edge times must be aggregated in *pace*, $t_e = \ell_e \sum_i t_i / \sum_i \ell_i$; averaging the segments' travel times instead made 26% of edges imply speeds above the network maximum and silently broke A*'s admissibility. Free-flow speeds are re-annotated from the generated network so numerator and denominator share one road model, which raises mean free-flow speed to 51.5 km/h: **absolute times here are optimistic and only ratios are reported**.

### B. Algorithms

Dijkstra [5] on the congested weight supplies the reference cost for every stretch figure; A* [6] uses a great-circle/max-speed heuristic, verified equal to Dijkstra rather than assumed so.

**MMAS** [2] uses $\tau^\alpha \eta^\beta$ with $\eta = 1/t_{uv}$ and bounded pheromone. Plain MMAS does not work here: with mean out-degree 2.46 the ant completion rate was *zero*, so we construct tours with backtracking plus a dead-node memo, raising completion from 36% to 80%. The memo is lossy by measurement (about 66% of prunes refuse a still-viable node), and we document it as a deliberate heuristic. A goal-directed

variant using A*'s heuristic is an ablation, not the default: it would give ACO target-location information Physarum never receives, and it is also not better.

**Physarum** carries conductivity $D_e$ and cost $L_e$ per edge, giving conductance $w_e = D_e/L_e$. Each iteration solves a weighted-Laplacian system for pressures (grounded at the sink), computes fluxes $Q_e = w_e \Delta p_e$, and adapts $D_e \leftarrow D_e + \mathrm{d}t\,(f(|Q_e|) - D_e)$ with $f(Q) = |Q|^\mu/(1 + |Q|^\mu)$ [4]. Convergence is $\max_e |\Delta D_e| < \epsilon$. **Pressures are solved on an undirected projection while an extracted path must be directed-legal**, which is the origin of most of what follows.

**Extraction, and its provenance.** The model yields a field, not a route. We extract in three stages and record which produced the reported path: (i) the *maximum-flux walk*, stepping to the highest-flux admissible successor with a visited set, the model's own extraction, procedurally identical to [12]; (ii) a shortest-path search on the $D_e > \theta$ subgraph if the walk dead-ends; (iii) a shortest-path search on the whole graph otherwise, which *is plain Dijkstra*. Recording provenance per run is what makes Section IV-A possible; without it a Physarum stretch of 1.000 is unfalsifiable.

Under warm start we zero severed edges' conductivity, drop them from the Laplacian and retain the rest; cold start resets $D$ to uniform. Closures remove edges but not nodes, so a stranded node yields a singular Laplacian and non-finite pressures, under which extraction silently falls through to the Dijkstra fallback. We restrict the Laplacian to the source's connected component in the surviving graph, and rejected an epsilon floor on $D$, which would invent transport capacity across a closed road.

### C. Design and statistics

Seven scenarios: no disruption (S0); one random path edge (S1); the highest-betweenness path edge (S2); three (S3) and five (S4) simultaneous path edges; a corridor (S5); and an intersection with all incident edges (S6). Closures are chosen relative to the current optimal path, must leave the pair connected, and rejected candidates are counted.

The matrix is 30 OD pairs (stratified by distance, disjoint from all tuning draws) $\times$ 7 scenarios $\times$ 2 demand levels $\times$ {warm, cold}: **5,520 runs**. ACO gets five seeds; Physarum one, justified by measurement (three initialisation seeds gave costs identical to four decimal places). All algorithms in a cell share one Dijkstra denominator.

Distributions are right-skewed and, for warm Physarum, bimodal, so we report **median and IQR, never means**, with paired Wilcoxon signed-rank tests [19] on identical (OD, scenario, level, seed) tuples and a two-stage clustered bootstrap (10,000 resamples) for intervals. Pre-disruption quality is deduplicated on the initial solve, which is identical across scenarios and start modes.

## IV. Results

The sweep completed 5,520 runs with 28 failures (0.51%), all one cause: two cells where no five path edges could be closed without disconnecting the pair. All stretch invariants hold with zero violations.

TABLE I
PROVENANCE OF PHYSARUM'S REPORTED PATH (776 SUCCESSFUL ROWS). THE MEDIAN STRETCH IS 1.000 IN *every* CLASS.

| Extraction method | Own? | $n$ | Share | Exact |
|---|---|---|---|---|
| `full_graph_fallback` | no | 315 | 40.6% | 95.2% |
| `max_flux_walk` | yes | 232 | 29.9% | 74.6% |
| `theta_subgraph_fallback` | no | 229 | 29.5% | 94.3% |

### A. Extraction provenance decides what a Physarum number means

Pooled across extraction methods, Physarum matches the optimal route on **90.0%** of undisrupted pairs at heavy demand and 86.7% at light, reproducing the accuracy this model is credited with. Restricted to its own maximum-flux walk, the same solves give **23.3% and 16.7%**. The walk produces any path at all on 69.8% of Physarum rows.

About **70% of reported Physarum paths did not come from its own extraction**, and `full_graph_fallback` is plain Dijkstra. Since the median stretch is exactly 1.000 within each class (Table I), a pooled median cannot separate "the model found the optimum" from "the fallback returned Dijkstra's answer", and a pooled Physarum-versus-Dijkstra comparison is partly circular.

On recovery the effect is sharper and *mode-dependent*. Cold start splits walk/threshold/full-graph as 95/201/62 (17.3% Dijkstra); warm start as 119/**0**/**239**. With roughly 98.8% of retained tubes below $\theta$, warm's threshold subgraph never connects the pair, a hard zero across all 358 cells, so **warm Physarum reports plain Dijkstra's path on two-thirds of recoveries**.

Fig. 1 is the clearest evidence: in two of three panels the with-fallback curve sits at stretch 1.0 from iteration one, because it was Dijkstra's answer before the model did any work. We therefore report Physarum as two labelled series throughout and never pool them.

For context, ACO reaches 43.3%/44.0% exact match with median stretch 1.0052 (light) and 1.0209 (heavy), and Dijkstra and A* are exact by construction. ACO *misses* the exit criterion we set in advance (stretch $\leq 1.10$ on $\geq 90\%$ of pairs), reaching 73.3% under free flow with a 95% CI of [58.7, 86.7] entirely below the bar; the failure mode is a colony settling on a wrong corridor, with MMAS's own convergence criterion firing on most such solves. We report this rather than retuning until it passes.

### B. Warm start buys speed and charges quality

Table II gives the raw cost, and the first thing it shows is that **Dijkstra wins on wall-clock, which we report rather than hide**: a recompute on this graph costs about 31 ms where the bio-inspired methods take seconds. The defensible quantities are iterations-to-adapt and post-disruption quality.

Paired on identical cells, warm start cuts Physarum's recovery by **2.97$\times$ in iterations** (faster on 350/358, $p = 3.2 \times 10^{-59}$) and **6.87$\times$ in wall-clock** (336/358, $p = 9.0 \times 10^{-52}$); ACO gains 1.26$\times$ and 1.56$\times$ ($n = 1{,}790$).

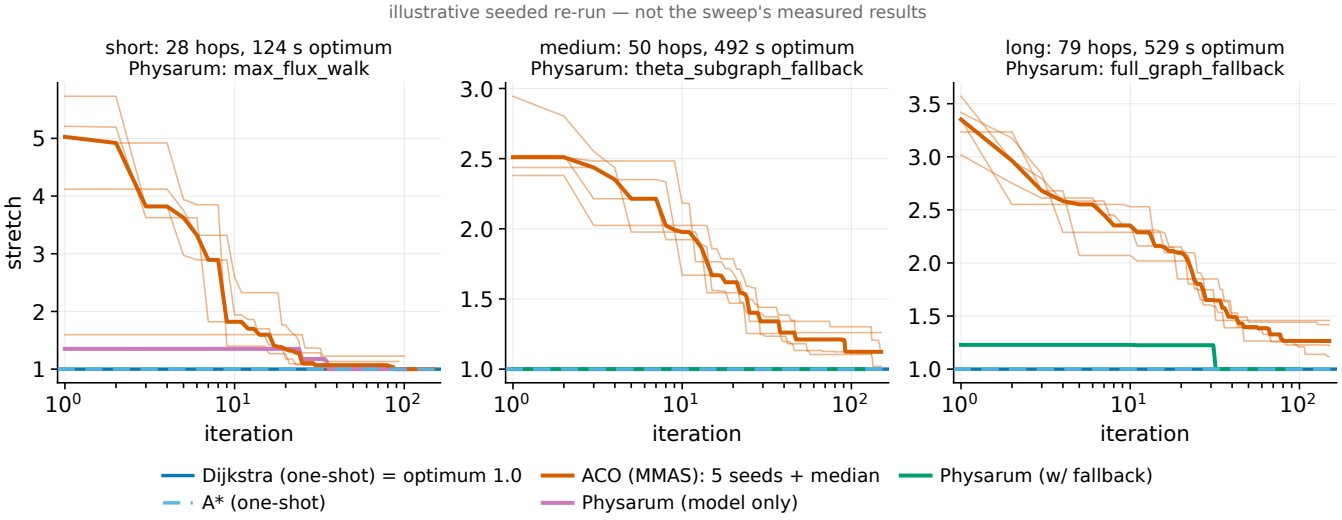



Fig. 1. Cost versus iteration across three distance strata. The Physarum with-fallback series starts at stretch 1.0 in two of three panels: that is the fallback, not convergence.

TABLE II
RECOVERY COST, SCENARIOS S1–S6. DIJKSTRA IS THE HONEST BASELINE: NO METHOD HERE APPROACHES IT ON WALL-CLOCK.

| Algorithm | Start | $n$ | Iters | Wall (s) | 1 iter |
|---|---|---|---|---|---|
| Dijkstra | cold | 358 | 1.0 | 0.031 | 100% |
| A* | cold | 358 | 1.0 | 0.027 | 100% |
| ACO | warm | 1790 | 69.0 | 34.07 | 0% |
| ACO | cold | 1790 | 110.0 | 59.80 | 0% |
| Physarum | warm | 358 | 42.0 | 0.235 | **48.3%** |
| Physarum | cold | 358 | 98.0 | 5.661 | 0% |

Two points matter more than the ratios. The distribution is **bimodal, not merely shifted**: 48.3% of warm recoveries converge in exactly one iteration while cold start never does and its minimum is 73 (Fig. 2). The ratio of the two medians is therefore not a valid statistic; an earlier version of this analysis computed one and produced a spurious 24×. And **per-iteration cost is essentially identical** warm versus cold (0.0511 vs 0.0538 s), establishing that the saving is fewer iterations rather than cheaper ones.

**The cost.** On the reported path, paired across all 358 cells, cold start is better: cold wins 57, warm 11, 290 ties ($p = 4.3 \times 10^{-7}$). On the model's own extraction, the warm walk completes on only 36.3% of recoveries against cold's 59.5%. Reporting the speed-up alone describes half the effect.

**Why.** We tested the obvious mechanism, a retained field driving flux down severed corridors, and *refuted* it: a placebo closure at least eight hops away reproduces nearly the whole deficit and is indistinguishable from a closure directly on the route (43.3% vs 41.7%, $n = 60$ paired, $p = 1.00$). A second candidate, that warm start leaves less conductivity, is *analytically impossible*: scaling $D$ by $c$ scales $w$ by $c$ and $p$ by $1/c$, leaving $Q = w\Delta p$ unchanged (verified numerically to $< 10^{-15}$). What the data support is that **the walk works on a near-uniform field and stops working as the field converges onto a thin corridor**: convergence alone, with no closure, drops completion from 69.8% to 43.0%, and cold start's advantage is that it rewinds the field to a walkable state.

### C. What adaptation contributes

This invites the conclusion that the model should stop earlier. Against a zero-adaptation control (the same pressure solve on a uniform field, containing none of the Tero dynamics, and exact by construction since the solver extracts before updating $D$), the converged field completes *less* often (0.433 vs 0.567) yet is far better when it does (median stretch **1.0059** vs 1.4367; exact given completion 46.2% vs 5.9%). Paired, the unadapted field loses 26–0 to the best adapted iteration ($n = 34$, $p = 8.3 \times 10^{-6}$), reproducing at scale on cold recovery (120–1, $p = 3.2 \times 10^{-21}$).

**Completion and quality move in opposite directions**, because a near-uniform field is walkable precisely *because* it is uninformative. This also resolves the tension with [12]: their claim concerns quality, which does improve with convergence; what fails here is completion, which a complete graph cannot exhibit. Consequently **no earlier stopping rule is worth adopting**: a perfect per-cell oracle buys *zero* additional exact matches, since every exact-match walk is already exact at convergence, and the router already retains the best walk across iterations.

### D. The congestion model decides the sign of the effect

On the undisrupted solve, ACO's within-10% rate is 81.3% under light SUMO demand and 58.0% under heavy, and paired on the same (OD, seed) cells heavy is significantly worse ($p = 0.036$; 37 pairs pass only under light against 2 only under heavy).

This *reverses* what we measure on the analytic BPR provider, where congestion improves ACO's outcomes because it flattens the cost surface so more routes land within 10% of

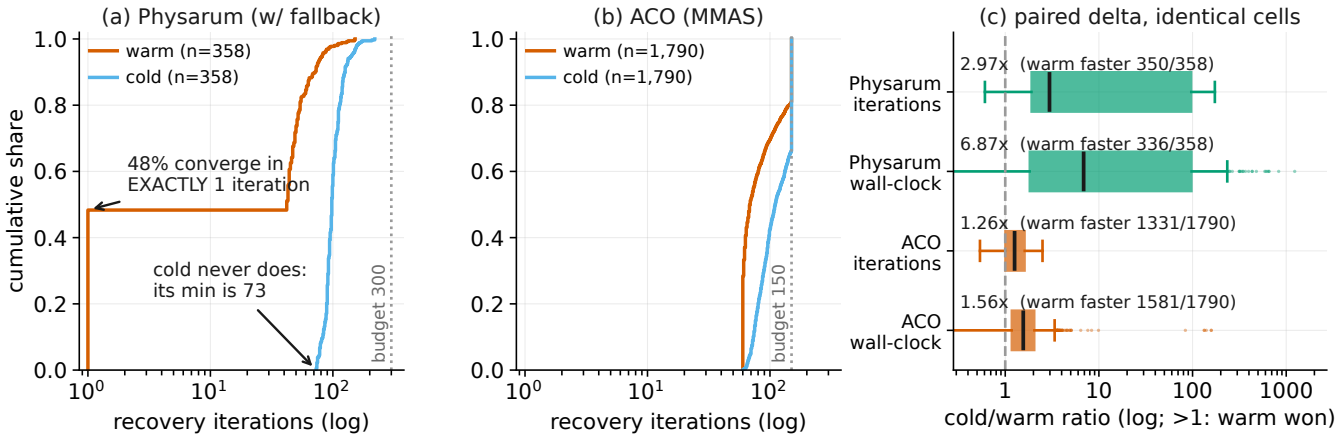



Fig. 2. Warm versus cold recovery. The mode at exactly one iteration is the shape of the result and the reason we report paired ratios.

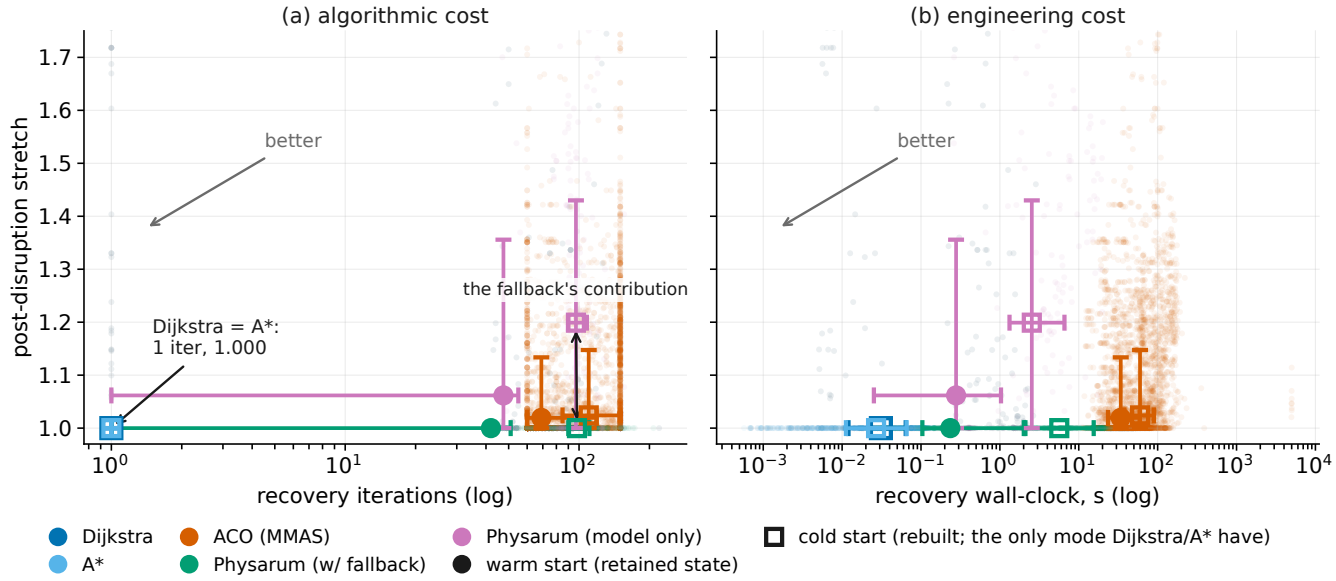



Fig. 3. The speed-versus-quality trade. Markers are medians with IQR arms over the individual S1–S6 recovery runs; Dijkstra and A* anchor the lower left. The model-only and pooled Physarum series share the same runs, so *the vertical gap between them is the fallback's contribution*.

optimal. The algorithm, the network and the OD pairs are identical; only the congestion source differs. SUMO puts signal delay in the numerator but not the free-flow denominator, while BPR loads synthetic demand onto shortest paths, which is exactly where an OD-path measurement walks.

**So the sign of the congestion effect is a property of the congestion model, not of the algorithm**, and any claim about routing behaviour "under congestion" that does not name its congestion model is underdetermined. We accordingly retract a framing from our own earlier analysis, that the two bio-inspired methods fail in opposite directions under congestion, which held on BPR and does not survive SUMO, where both degrade.

## V. Discussion

**Why a real road network was necessary.** The maximum-flux walk fails by *dead-ending*: reaching a node with no admissible unvisited successor. On a complete graph that is impossible, and on a small-world of mean degree 6–600 it is vanishingly rare. Our network has mean out-degree 2.46, and the undirected pressure solve routinely prefers a direction the road forbids, so the failure mode we measure is structurally unreachable on the substrates where this model's accuracy was established. That reframes the disagreement with prior work as a matter of domain rather than correctness: neither [11] nor [12] is wrong on its own graphs. What changes on a directed, sparse, organically grown network is that extraction stops being a formality and becomes the step that decides the reported outcome.

**Provenance as a reporting requirement.** A Physarum accuracy figure on such a network is a statement about the extraction policy at least as much as about the model, and a pooled number is unfalsifiable, since our 90% and our 23.3% come from the same solves. Any evaluation of this model class on a directed network should publish the provenance distribution of its reported paths, and a model whose fallback is a shortest-path algorithm should report what share of its results that fallback produced. The stakes are concrete: the stranded-component degeneracy of Section III produced a fabricated mean stretch of exactly 1.000 on the scenario where the model is weakest, with no error raised, and provenance instrumentation is what made it visible.

**What self-healing is worth here.** The adaptive state is real and substantial: single-iteration recovery on nearly half of all closures, at unchanged per-iteration cost, which is the strongest form of the claim we can support. But it is paid for in path quality, on both the reported path and the model's own extraction. A system that must guarantee route quality after a closure should prefer a cold solve or a recompute; one that must respond within a single adaptation step and can tolerate a few percent of path cost is where retained state pays. Neither conclusion follows from the speed measurement alone. And the baseline stands: Dijkstra recomputes an exact route in about 31 ms here, so at this network size recomputation is simply the right engineering answer, and the case for adaptive methods must rest on regimes we did not test: far larger networks, many simultaneous commodities, or settings where the cost function is not globally observable.

**Limitations.** One city and one bounding box, so the structural properties we argue are decisive, while common to dense organically grown cities, are not demonstrated to transfer. Congestion comes from synthetic demand rather than calibrated counts, which is where the closest recent ACO work is stronger [14]; we traded measured demand for a controlled, repeatable closure protocol over a full directed street network. Absolute times are optimistic because the shared baseline re-annotates free-flow speeds, so we report only ratios. Physarum is run per OD pair rather than as multi-commodity flow, and we did not implement the directed formulation [10], so our findings characterise the canonical model on a directed network, an acknowledged limitation of that model, rather than the limits of Physarum-style methods in principle. ACO was tuned while Physarum was near-default, a real asymmetry, though the extraction findings concern a structural failure mode rather than a parameter setting.

## VI. Conclusion

Benchmarking classical, ant-colony and Physarum routing on a real directed road network under microsimulated congestion, over 5,520 paired runs, we find that the self-healing claim holds in its strongest form, and that it is a trade: warm start recovers in a single iteration on 48.3% of closures, which cold start never achieves and at unchanged per-iteration cost, yet is significantly worse on the reported post-disruption path. The finding we think matters most is methodological. A Physarum model emits a field, not a route, and on a sparse directed network the extraction step decides the answer: pooled, our Physarum matches the optimum on 90% of pairs; using only its own walk, 23.3%. Because the median stretch is exactly 1.000 inside every provenance class, a pooled figure cannot be falsified, so we argue provenance should be reported as a matter of course. Throughout, Dijkstra remained the honest baseline that no method here approached on wall-clock.

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